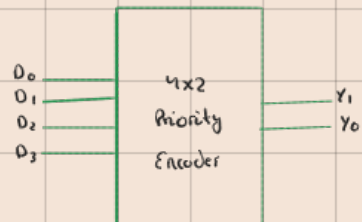


Qno 1

Truth Table:



D_3	D_2	D_1	D_0	Y_1	Y_0
1	0	0	0	0	0
x	1	0	0	0	1
x	x	1	0	1	0
x	x	x	1	1	1

Equations:

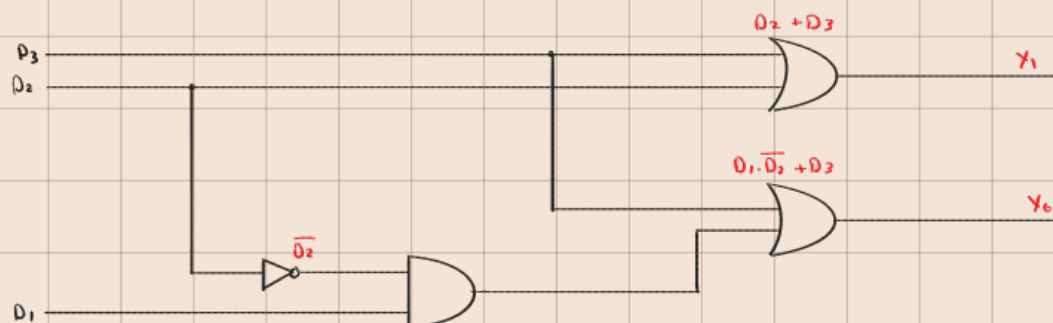
$$Y_0 = \overline{D_3} \cdot \overline{D_2} \cdot D_1 + D_3$$

$$Y_1 = \overline{D_3} \cdot D_2 + D_3$$

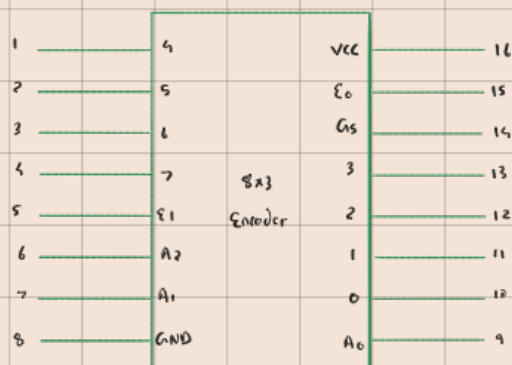
$$Y_0 = D_1 \cdot \overline{D_2} + D_3$$

$$Y_1 = D_2 + D_3$$

Logic Diagram:



Qno 2



- It is a 8 to 3 line Priority Encoder which takes 3 individual inputs and gives out a 3 Bit Binary Output.
- The Priority feature in this IC works when multiple Input Pins are active at the same time. The IC encodes the Input with the highest numerical Value.
e.g: Input 3 and 6 are both active, - the output will be 6
- This IC is mostly Used in interrupt Controllers and keyboard Encoders.

Pins and their Uses:

- This IC usually comes in 16 pins and uses Active Low Logic
(Signals are considered ON with a low Input "0")

Pins: [10-13, 1-4] Inputs

Inputs: 0-3, 4-7

Eight Active low Inputs representing the

Inputs from 0₁₀ to 0₇ of the Decimed Values to be Encoded.

Pins: [9, 6, 7]

Outputs

Outputs: A₀, A₁, A₂

A 3 bit Inverted binary code of the highest priority Active Input.

Pins: [5]	Enable Input	Must be Low for the IC to operate, If High all outputs are disabled (High)
Enable pin E1		

Pins: [15]		
Enable pin E0	Enable Output	Goes low Only if E1 is low and no data inputs (0-7) are active. Used in Cascading the IC

Pins: [14]	Group Signal	Goes low when at least one data Input is active. Acts as an Interrupt Flag
GS		

Pins: [16, 8]	Power Pins	Power Supply Positive and Negative.
VCC / GND		

Functionality:

- It is a 8 to 3 line Priority Encoder.
- It consists of 8 Individual Input lines (D_0 - D_7) and 3 Output lines (A_0 , A_1 , A_2)
- All inputs and outputs are considered Active Low.
- This IC converts the highest priority active Input (Decimal Value) into a 3 bit Binary Code.
- All the Inputs have fixed priority
$$D_7 > D_6 > D_5 > D_4 > D_3 > D_2 > D_1 > D_0$$
- When multiple Inputs are low at the same time the Decimal Value of the highest priority Input is converted into a 3-Bit Binary Output.

Truth Table:

ε_1	D_7	D_6	D_5	D_4	D_3	D_2	D_1	D_0	A_0	A_1	A_2
1	x	x	x	x	x	x	x	x	1	1	1
0	0	x	x	x	x	x	x	x	0	0	0
0	1	0	x	x	x	x	x	x	0	0	1
0	1	1	0	x	x	x	x	x	0	1	0
0	1	1	1	0	x	x	x	x	0	1	1
0	1	1	1	1	0	x	x	x	1	0	0
0	1	1	1	1	1	0	x	x	1	0	1
0	1	1	1	1	1	1	0	x	1	1	0
0	1	1	1	1	1	1	1	0	1	1	1

Equations:

$$A_0 = \overline{I_7 + I_6 + I_5 + I_4}$$

- A_0 is 0 when these 4 inputs are 0

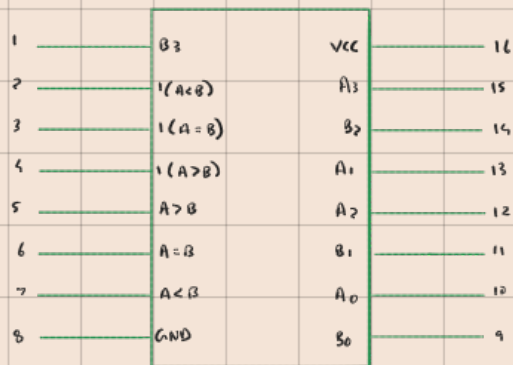
$$A_1 = \overline{I_7 + I_6 + I_3 + I_2}$$

- A_1 is 0 when these 4 inputs are 0

$$A_2 = \overline{I_1 + I_3 + I_5 + I_7}$$

- A_2 is 0 when these 4 inputs are 0

Qno 3



7485

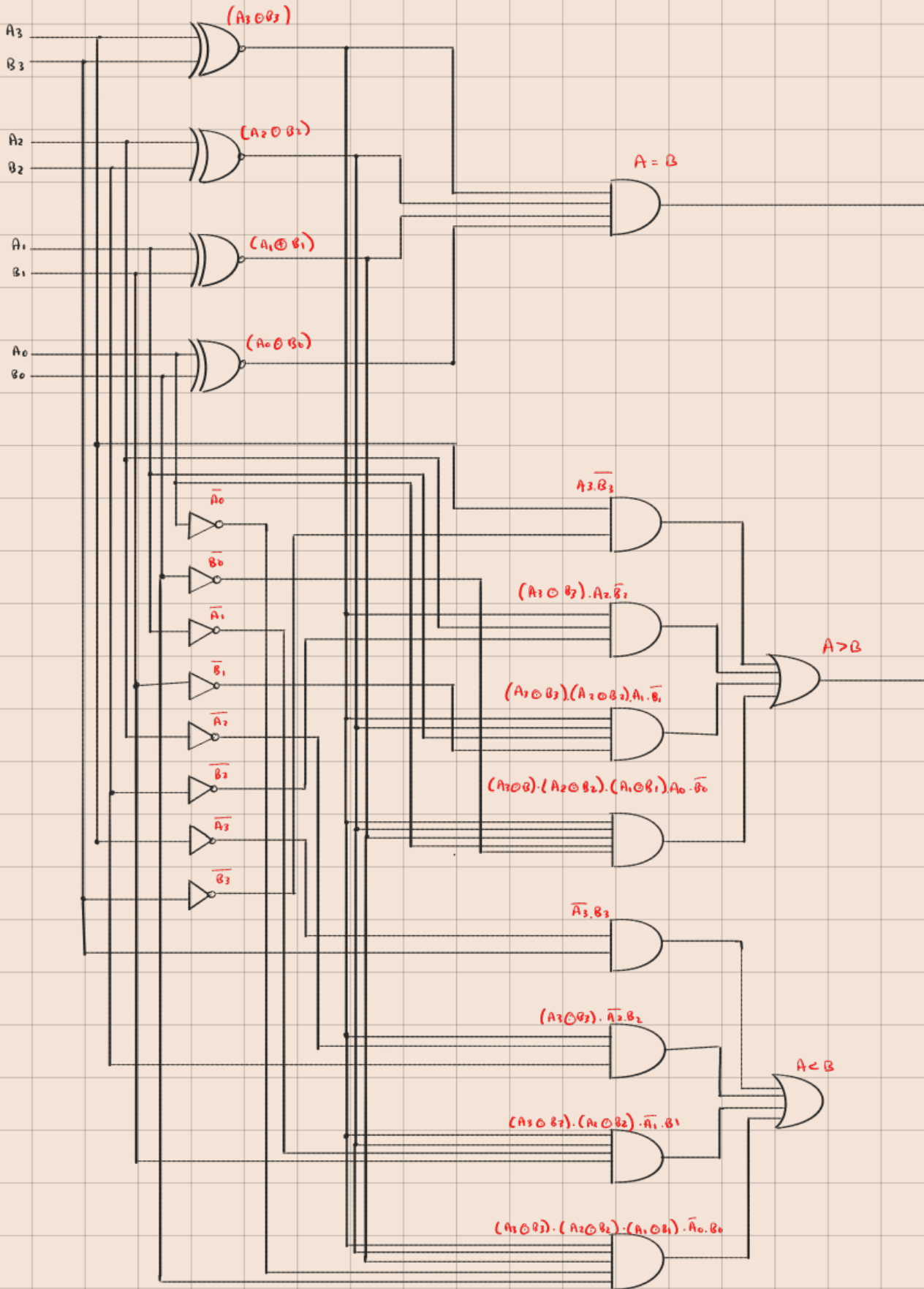
4 bit comparator

First it compares the Most Significant Bits, the A_3 and B_3 inputs. If equal it moves onto A_2 and B_2 , then A_1 and B_1 . if equal it finally compares A_0 and B_0 . If all the 4 bits are equal then $A = B$.

Truth Table:

$A_3 B_3$	$A_2 B_2$	$A_1 B_1$	$A_0 B_0$	$A > B$	$A = B$	$A < B$
$A_3 > B_3$	X	X	X	1	0	0
$A_3 < B_3$	X	X	X	0	0	1
$A_3 = B_3$	$A_2 > B_2$	X	X	1	0	0
$A_3 = B_3$	$A_2 < B_2$	X	X	0	0	1
$A_3 = B_3$	$A_2 = B_2$	$A_1 > B_1$	X	1	0	0
$A_3 = B_3$	$A_2 = B_2$	$A_1 < B_1$	X	0	0	1
$A_3 = B_3$	$A_2 = B_2$	$A_1 = B_1$	$A_0 > B_0$	1	0	0
$A_3 = B_3$	$A_2 = B_2$	$A_1 = B_1$	$A_0 < B_0$	0	0	1
$A_3 = B_3$	$A_2 = B_2$	$A_1 = B_1$	$A_0 = B_0$	0	1	0

Logic Diagram:



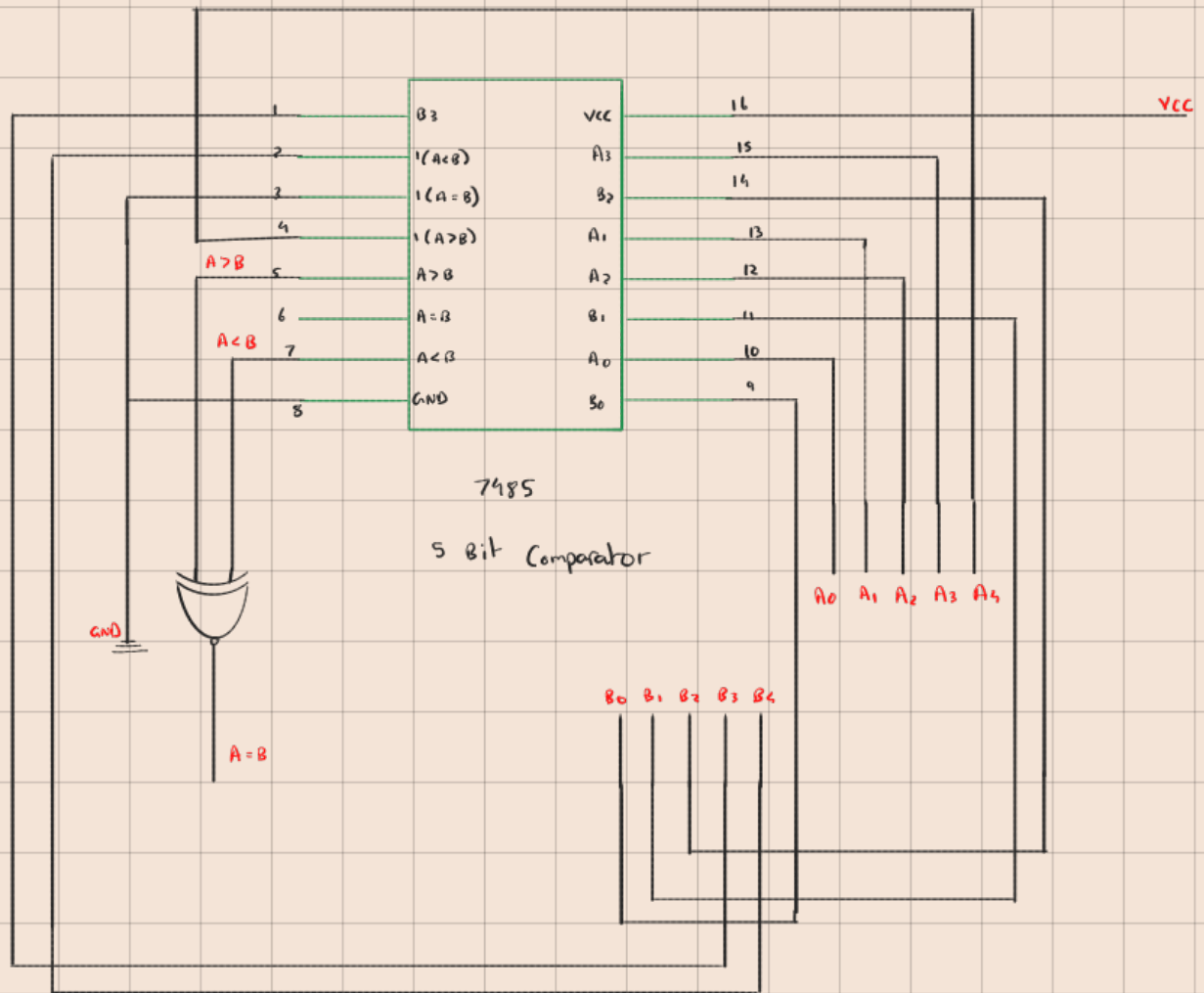
Equations:

$$A = B : (A_3 \odot B_3) \cdot (A_2 \odot B_2) \cdot (A_1 \odot B_1) \cdot (A_0 \odot B_0)$$

$$A > B : A_3 \bar{B}_3 + (A_3 \odot B_3) \cdot A_2 \bar{B}_2 + (A_3 \odot B_3) \cdot (A_2 \odot B_2) \cdot A_1 \bar{B}_1 + (A_3 \odot B_3) \cdot (A_2 \odot B_2) \cdot (A_1 \odot B_1) \cdot A_0 \bar{B}_0$$

$$A < B : \bar{A}_3 B_3 + (A_3 \odot B_3) \cdot \bar{A}_2 B_2 + (A_3 \odot B_3) \cdot (A_2 \odot B_2) \cdot \bar{A}_1 B_1 + (A_3 \odot B_3) \cdot (A_2 \odot B_2) \cdot (A_1 \odot B_1) \cdot \bar{A}_0 B_0$$

Qno 4



Inputs	Pin#
A ₀	10
A ₁	13
A ₂	12
A ₃	15
A ₄	4
B ₀	9
B ₁	11
B ₂	14
B ₃	1
B ₄	2

output	pin #
$A < B$	7
$A > B$	5
$A = B$	$(7 \oplus 5)$